CS-UH 2012 - Software Engineering-004 **Name:** ______________________
Spring 2025
Midterm Exam **Net ID:** ______________________
Time Limit: 60 minutes

This exam contains 11 pages (including this cover page) and 19 questions.
Total of points is 40.

Grade Table (for Professor use only)

Question	Points	Score
1	1	
2	1	
3	1	
4	1	
5	1	
6	1	
7	1	
8	1	
9	1	
10	1	
11	1	
12	1	
13	1	
14	1	
15	1	
16	3	
17	4	
18	8	
19	10	
Total:	40	

Part 1: Multiple Choice Questions

1. (1 point) In UML, aggregation is typically represented by which symbol?
   - ◯ A solid diamond
   - ◯ A hollow diamond
   - ◯ A solid arrow
   - ◯ A dashed line

2. (1 point) In a system where many objects require support for logging, rather than adding logging responsibilities to each class, which GRASP principle should be applied?
   - ◯ Expert.
   - ◯ Creator.
   - ◯ Pure Fabrication.
   - ◯ Don't Talk to Strangers.

3. (1 point) At the beginning of a sprint, the team holds a sprint planning meeting. What is the main objective of this meeting?
   - ◯ To review past performance metrics.
   - ◯ To collaboratively review the product backlog and commit to a set of items that the team can complete during the sprint.
   - ◯ To elicitate new requirements and add these to the product backlog.
   - ◯ To document detailed design contracts.

4. (1 point) You're using GRASP principles to design a system using the object- oriented approach. There is a grow() method that is used by a Tree class and can be used in different ways by Maple and Palm. What pattern of GRASP does this scenario represent?
   - ◯ Creator.
   - ◯ Polymorphism.
   - ◯ Pure fabrication.
   - ◯ Controller.

5. (1 point) Which of the following is an example of a high-level refactoring technique?
   - ◯ Class hierarchy restructuring.
   - ◯ Consolidate duplicate code into a common method.
   - ◯ Renaming variables for clarity.
   - ◯ adding comments to variables to reflect their usages.

6. (1 point) A POST class needs to obtain the current payment amount from a Sale, which in turn holds a Payment object. Which design solution minimizes coupling according to the "Don't Talk to Strangers" principle?
   - ◯ The POST class directly accesses the Payment object's `amountTendered()` method.
   - ◯ The POST class asks the Sale class to provide the payment amount through a dedicated method.
   - ◯ The POST class creates a new Payment object to calculate the amount.
   - ◯ The Payment class exposes all its internal details publicly.

7. (1 point) A volunteer management system tracks Volunteers and Events. Each Volunteer may participate in multiple Events, and each Event can involve many Volunteers. Moreover, the participation record must capture details such as the volunteer's role (e.g., coordinator, helper), as well as the start and optional end dates of their involvement in each event. How should the **participation** be best represented in the domain model?
   - ◯ Model **Participation** as an association class between Volunteer and Event.
   - ◯ Represent **Participation** as an attribute within the Volunteer class.
   - ◯ Represent **Participation** as an attribute within the Event class.
   - ◯ Model **Participation** as a subclass of Event.

8. (1 point) In Scrum, who is responsible for removing challenges that hinder the team's progress?
   - ◯ Scrum Master.
   - ◯ Product Owner.
   - ◯ Software Tester.
   - ◯ Software Developer.

9. (1 point) Consider the following git command:

```
git commit -m "Add new feature for user authentication"
```

   Which of the following statements is correct?
   - ◯ This command pushes the changes to a remote repository with a commit message, since Git is a centralized version control system.
   - ◯ This command stages all changes in the working directory, preparing them to be saved in the next commit.
   - ◯ This command saves the changes to the local repository with a commit message, since Git is a decentralized version control system.
   - ◯ This command pulls the latest changes from a remote repository.

10. (1 point) A hotel reservation system has a "`Room`" class and a "`Reservation`" class. The "`Reservation`" class needs to track the check-out date for each room booked. How should this be modeled?

    ◯ As an attribute of "`Room`".

    ◯ As an attribute of "`Reservation`".

    ◯ As a separate "`CheckOutDate`" class.

    ◯ As a generalization of "`Room`".

11. (1 point) In the interaction diagram for the flight reservation system provided below, if we translate this design into code, which class is responsible for implementing the **`findFlights()`** method?

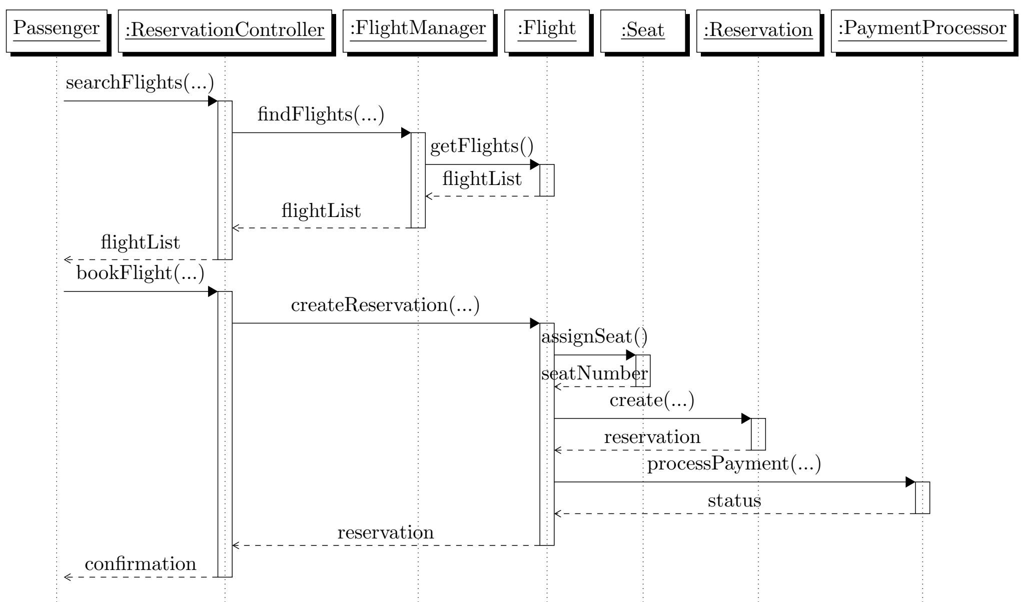

Figure 1: UML Sequence Diagram for Flight Reservation System

    ◯ ReservationController.

    ◯ FlightManager.

    ◯ Flight.

    ◯ Seat.

12. (1 point) In the above diagram, which class is responsible on calling the constructor method of `Reservation` class?

    ◯ ReservationController.

    ◯ Flight.

    ◯ Reservation.

    ◯ FlightManager.

13. (1 point) In a holiday booking system, the system allows users to book flights, hotels, and cruises. For each type of booking (flight, hotel, or cruise), the system prompts the user for payment information. Which use case relationship should be used to represent this shared behavior of collecting payment information?

    - ◯ <<include>>
    - ◯ <<extend>>
    - ◯ <<generalize>>
    - ◯ <<specialize>>

14. (1 point) A requirement reads: "The system shall process payments securely and quickly." What's the fix per IEEE 29148?

    - ◯ Split into "process securely with AES-256" and "process within 1 second".
    - ◯ Rewrite as "process payments using encryption quickly".
    - ◯ Add "if possible" for flexibility.
    - ◯ Keep combined with "under high load".

15. (1 point) In a university management system, a class diagram shows `"Professor"` inheriting from `"Faculty"`. Which UML notation correctly indicates this generalization?

    - ◯ A solid line with a hollow triangle at the `"Faculty"` end.
    - ◯ A dashed line with an open arrowhead at the `"Professor"` end.
    - ◯ A solid line with a diamond at the `"Faculty"` end.
    - ◯ A dotted line with a filled triangle at the `"Professor"` end.

Essay / Design Questions

16. (3 points) A software company is tasked with developing a new payroll system for a multinational corporation. How should the team decide between the Waterfall model and Agile methodologies? What factors would influence their decision?

17. (4 points) You are designing a use case for a **Smart Healthcare Monitoring System**. The use case is called **”Optimize Patient Health Monitoring”**. Below is the **Main Success Scenario** for this use case. Your task is to **Draw the System Sequence Diagram (SSD)** for the below **Main Success Scenario**.

Main Success Scenario: Optimize Patient Health Monitoring

1. The use case begins when the **Healthcare Provider** activates the ”Optimize Monitoring” mode via the healthcare management app.
2. The system retrieves real-time patient health data (e.g., heart rate, blood pressure, oxygen levels) from the **Wearable Devices**.
3. The system retrieves environmental data (e.g., temperature, air quality) from the **Environmental Monitoring Service**.
4. The system analyzes the patient’s historical health data and preferences (e.g., medication schedules, activity levels).
5. The system calculates the optimal monitoring plan to ensure patient safety and health improvement.
6. The system adjusts the settings of **Monitoring Devices** (e.g., frequency of data collection, alerts) based on the optimized plan.
7. The system notifies the **Healthcare Provider** of the adjustments made and the expected health outcomes.
8. The use case ends successfully.

18. (8 points) The below sequence diagrams illustrate the design of the two operations `trackEnergyUsage()` and `generateSustainabilityReport()` within the sustainability management system. Both operations belong to the same use case: "Monitor Sustainability Metrics". Please note that the `User` represents an actor in the system.

    Sketch the corresponding class diagram (one diagram), including the classes, associations, and methods as depicted in the two interaction diagrams.

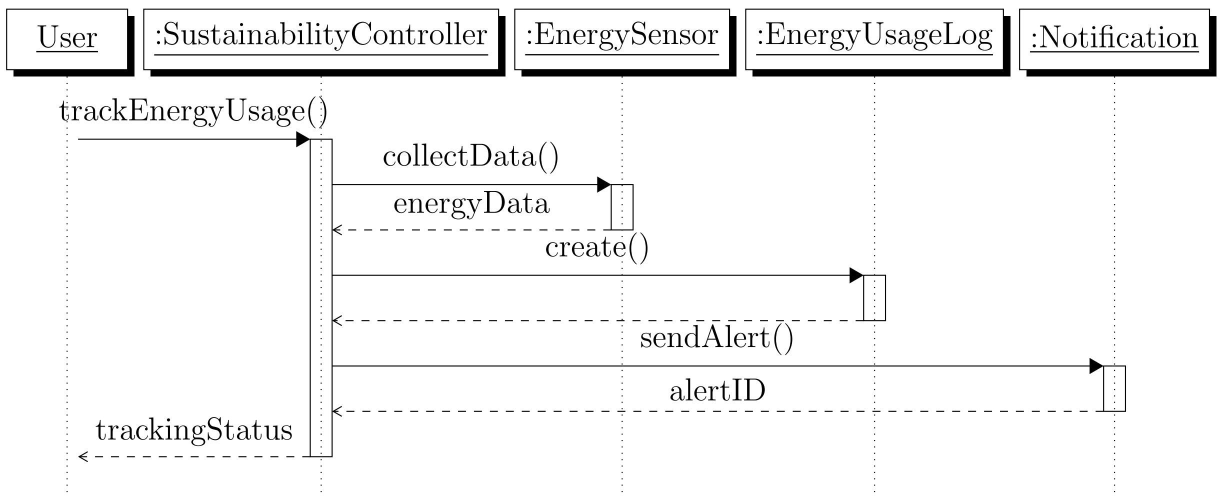

Figure 2: Sequence Diagram: trackEnergyUsage()

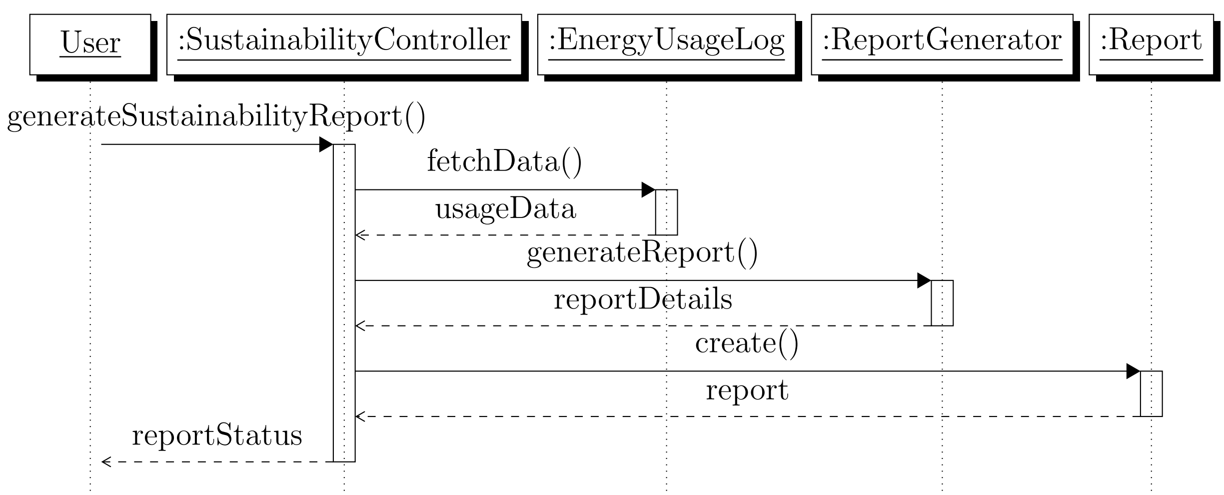

Figure 3: Sequence Diagram: generateSustainabilityReport()

19. (10 points) In the **Online Shopping System**, the two interaction diagrams below illustrate the interactions among objects when performing the operations `makePayPalPayment()` and `makeGiftCardPayment()` within the **Process Payment** use case. Please note that the `Customer` represents an actor in the system.

    List all GRASP patterns you can detect that are influencing the design of these two interaction diagrams. For each GRASP pattern you detect, clearly mention how this pattern is being represented in the interaction diagrams.

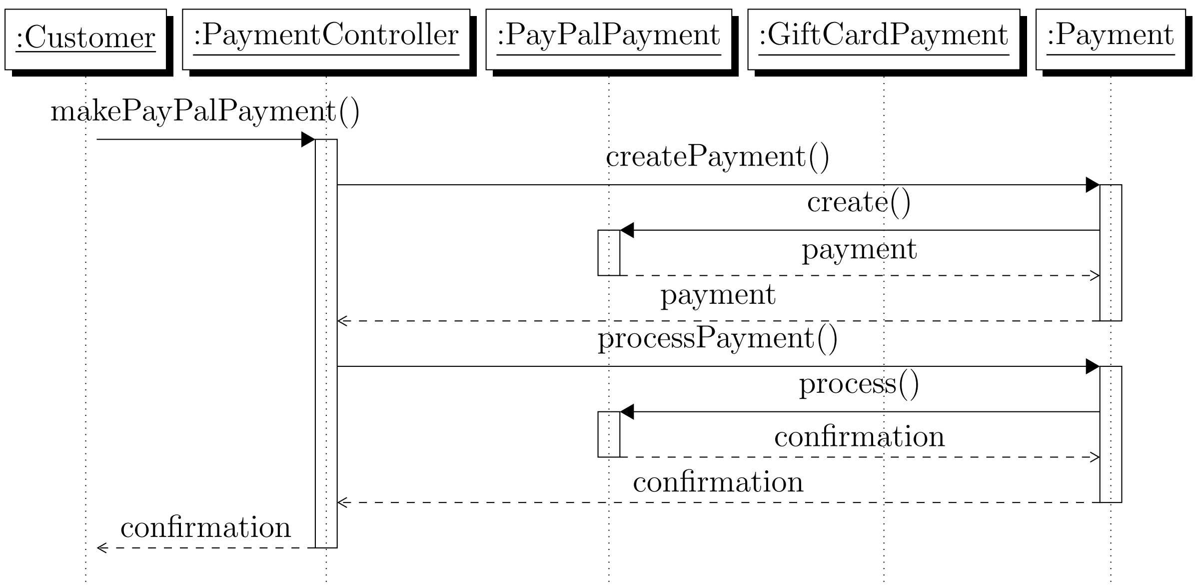

Figure 4: Interaction Diagram for `makePayPalPayment()`

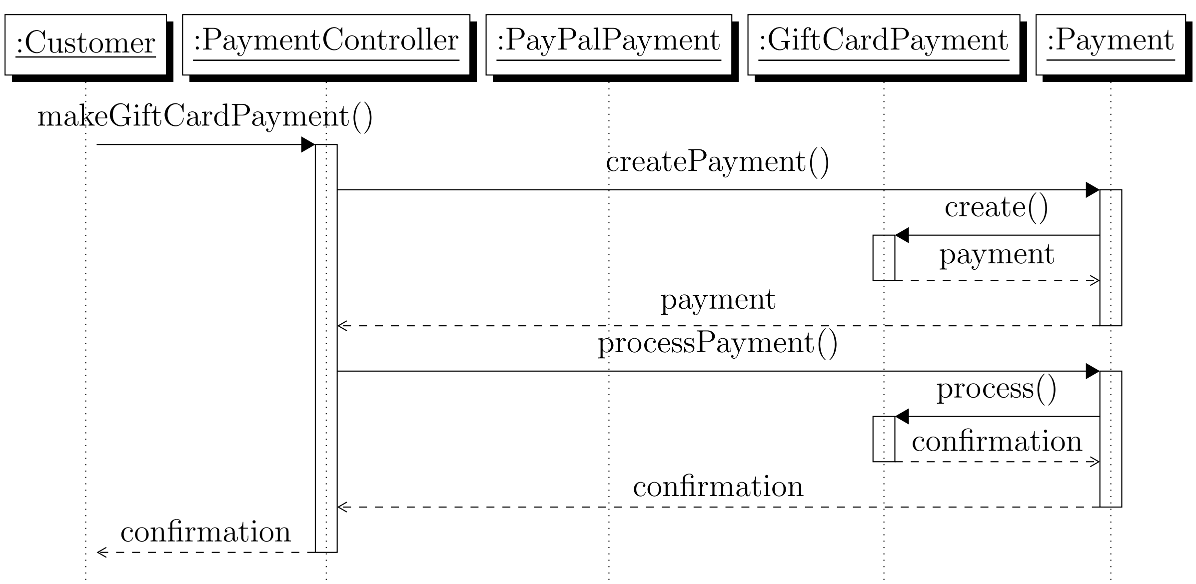

Figure 5: Interaction Diagram for `makeGiftCardPayment()`